AB CLOUD QUANTUM RESEARCH · MONOGRAPH SERIES

VOLUME TS1

TS1. The phase-cycle theorem

Supplementary study 'the machine with no long-term memory': three spinor phase models θ = π/2 · π/3 · π/7 — volatile state only, real-time steps, zero storage

Исаев Исхак Хамзатович

Независимый исследователь, г. Нальчик, Кабардино-Балкарская Республика

Исаев Исхак Хамзатович · version 1.0.0 · 2026-10-11

TS1. The phase-cycle theorem

Supplementary study 'the machine with no long-term memory': three spinor phase models θ = π/2 · π/3 · π/7 — volatile state only, real-time steps, zero storage

Исаев Исхак Хамзатович · Независимый исследователь, г. Нальчик, Кабардино-Балкарская Республика

AB Cloud Quantum Research · supplementary study · том TS1 · версия 1.0.0 · 2026-10-11

Contents

TS1. The phase-cycle theorem 2

Certificate ES1 (phase cycles) 2

Boundaries and honest limitations 3

TS1. The phase-cycle theorem

Theorem TS1 (ES1).

Let $\theta = \pi/k$, $k \in \mathbb{N}$, and $R: \varphi \mapsto (\varphi + \theta) \bmod 2\pi$ be a rotation of the circle S^1. Then: (i) the orbit of any point of the state lattice $G_\theta = \{0, \theta, 2\theta, \ldots\}$ closes after exactly $N = 2k$ steps (for the model phases: $\theta = \pi/2 \to N = 4$, $\theta = \pi/3 \to N = 6$, $\theta = \pi/7 \to N = 14$); (ii) the minimal angular gap between distinct states equals $2\pi/N = \pi/k$; (iii) the lattice is uniform: the points split the circle into N equal arcs. Corollary: the machine capacity is $\log_2 N$ bits per cycle (from 2 to 3.80735 bits/step for the model phases), while step and memory remain O(1).

> ***Proof.*** *(i) Closure: $N\cdot\theta = 2k\cdot(\pi/k) = 2\pi \equiv 0 \pmod{2\pi}$, hence $R^N = id$. Minimality: if $m\cdot\theta \equiv 0 \pmod{2\pi}$, then $m\pi/k = 2\pi j$ for some $j \in \mathbb{Z}$, whence $m = 2kj \geq 2k$ for $m > 0$ — so the minimal period is exactly $2k$. (ii) Adjacent lattice points differ by $\theta = \pi/k = 2\pi/(2k)$; for any $i \neq j$ the circular distance between $i\theta$ and $j\theta$ is a multiple of θ and at most π, so the minimal gap is $\theta = 2\pi/N$. (iii) The N points $\{i\theta\} = \{2\pi i/N\}$ are the roots of unity — uniform by construction.* ∎

Certificate ES1 (phase cycles)

Campaign ES1 ran with pre-registered thresholds; verdict: PASS.

All checks were fixed before the run; raw data (JSON + CSV) is committed under data/ and reproduced by the experiments/ scripts.

Table 1. ES1 certificate checks.

Check	Result	Threshold
C1 period = 2k	4/6/14 = periods exactly	= 2k
C2 min gap	matches 2π/N to 1e-9	±1e-9
C3 uniformity	≤ 8.9e-16 rad	≤ 1e-9
C4 серия k = 2…16	N(π/k) = 2k for all k	= 2k

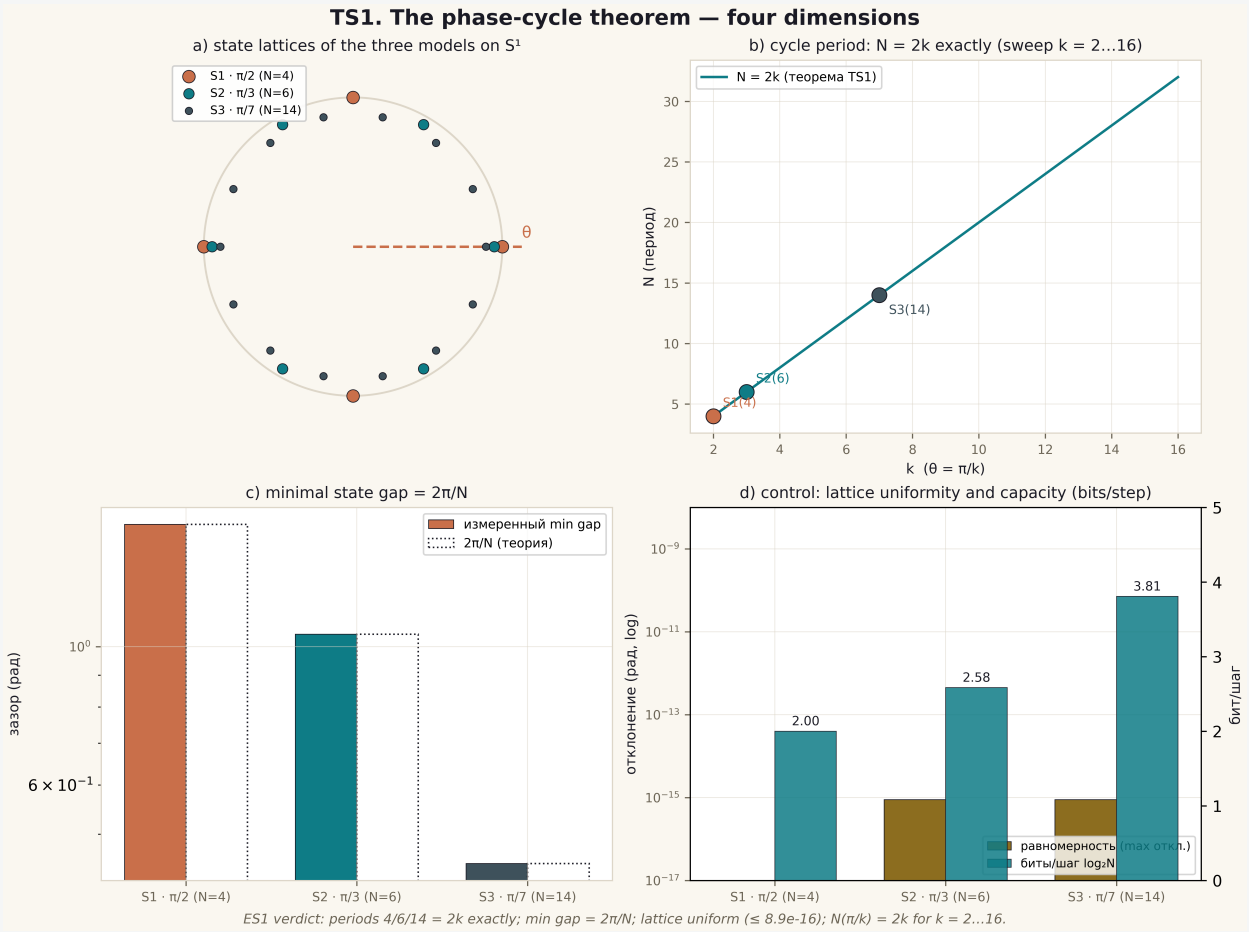


Fig. 1. Theorem TS1: four dimensions — state lattices, the period N = 2k, the minimal gap, the uniformity and capacity control (2×2 tile, 600 dpi, ES1 verdict data).

Boundaries and honest limitations

The theorem certifies closure of the phase lattices of the chosen models and the formula N = 2k; it says nothing about physical realisation of spinor phases or coherent carriers and is unrelated to the Riemann hypothesis. The capacity $\log_2 N$ grows logarithmically in k while the robustness gap shrinks as 1/k — the capacity-vs-robustness trade-off is published as an honest boundary (see ES5).